

Passing Arguments into a Function or Method

The arguments in a call to a function and the parameters defined in the function have a one-to-one relationship, so that the value of the first argument is given to the first parameter, the value of the second argument is given to the second parameter, and so on.

[Figure 7-1](#), duplicated from Chapter 3, shows that the same is true when you make a call to a method of an object. We can see that the first parameter, which is always `self`, is set to the object in the call.

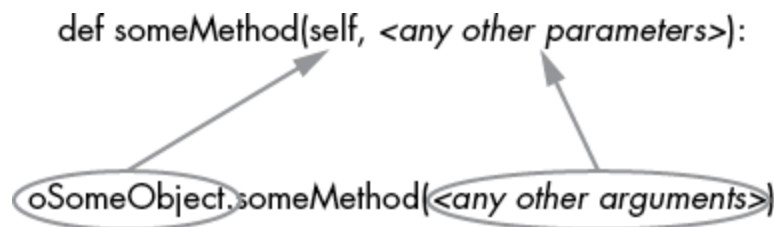


Figure 7-1: How arguments passed into a method match up with its parameters

However, Python (and some other languages) allows you to make some of the arguments optional. If an optional argument isn't provided in a call, we can provide a default value to use in the function or method instead. I'll explain by means of a real-world analogy.

If you order a hamburger at a Burger King restaurant, your burger will come with ketchup, mustard, and pickles. But Burger King is famous for saying, "You can have it your way." If you want some other combination of condiments, you must say what you want (or don't want) when you make your order.

We'll start by writing an `orderBurgers()` function that simulates making a burger order in the regular way we've been defining functions, without implementing default values:

```
def orderBurgers(nBurgers, ketchup, mustard, pickles):
```

You must specify the number of hamburgers you want to order, but ideally, if you want the defaults of `True` for adding ketchup, mustard, and